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Dear Editors ,

Herewith, we are submitting our manuscript “*Reconstruction of ancestral protein sequences using autoregressive generative models*” for consideration for publication in Molecular Biology and Evolution.

The field of generative protein models has recently seen major developments, with models that are now able to generate novel artificial proteins with tailored function. An essential ingredient of generative models is their ability to quantitatively describe epistasis: the interaction between amino acids at different sequence positions.

On the other hand, phylogenetic methods remain based on evolutionary models that consider sequence positions independently. This choice is not prompted by biological considerations but rather by computational feasibility: taking epistasis into account is considered as leading to overly complex models.

In this work, we show that a particular class of protein generative models based on the so-called *autoregressive* architecture can be used to describe the dynamics of sequence change in time. We apply this generative sequence evolution model to the task of ancestral protein reconstruction, showing that

- taking epistasis into account leads to more accurate reconstructions than traditional site-independent models;
- the generative evolution model can be used to sample high quality sequences at internal nodes of a tree in a Bayesian way, reducing biases due to maximum-likelihood estimates;
- the generative model efficiently handles insertions/deletions, reconstructing ancestral sequences that can have alignment gaps.

Our results are obtained using both *in silico* evolution of proteins and data obtained in experimental evolution studies. The use of such experimental data to assess the quality of ancestral reconstruction is itself novel.

Our manuscript attempts to bridge a gap between the blooming field of generative protein models and phylogenetic techniques. It also proposes robust ways of measuring the quality and bias in ancestral sequence reconstruction. For these reasons, we think that it is well suited for the readership of Molecular Biology and Evolution.

Sincerely,

Pierre Barrat-Charlaix, on behalf of all the authors